Chapter 10

Less invasive options

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Smaller incisions

Anterior MIDCAB

There are two main options: (1) 6–10cm incision from the left border of the sternum over the 4th rib which may be resected (Fig. 10.1a), or (2) a 5–7cm muscle splitting incision over the 4th rib space extending towards the nipple, from 2cm lateral to the left sternal edge. A Finochietto, West self-retaining or soft-tissue retractor is used. The pericardium is hitched to the chest wall with 1/0 silk sutures. Suitable for:

- MIDCAB (minimally invasive direct coronary artery bypass grafting).
- Endo ACAB (endoscopic atraumatic coronary artery bypass grafting).

Right minithoracotomy

A 6–10cm incision is made along the 4th rib space extending from the level of the nipple to the anterior axillary line, or following the breast crease (Fig. 10.1d). Pectoralis major is divided or split, and the intercostal muscle raised off the inferior rib anteriorly to 1cm lateral to the sternum (avoiding the RIMA) and laterally round to the costal angles to mobilize the rib space. A mini Cosgrove, McCarthy, or similar mitral retractor with narrow blades is placed between the ribs and the pericardium opened well above the right phrenic nerve. It is possible to cannulate the aorta, SVC, IVC, and coronary sinus directly through this, or peripherally in order to create more space (📖 p120). This incision is useful for:

- Mitral and tricuspid repair.
- ASD repair.

Hemisternotomy

A full sternotomy can be performed through a very low 6–10cm skin incision in most patients and gives excellent access for mitral and aortic surgery. A ***hemisternotomy*** is an alternative where only the upper or lower 2/3 of the sternum is divided. Its main disadvantage is the risk of tearing the IMA or intercostals when opening the retractor. A 6–10cm incision is made in the midline from an inch or so above the xiphisternum to an inch or so below the angle of Louis. The sternum is divided from xiphisternum to the 3rd or 4th intercostal space where it is 't'd' to the right (***lower hemisternotomy***) or from the sternal notch down to the 3rd or 4th intercostal space (***upper hemisternotomy***) (Fig. 10.1b). The tissue over the 4th rib space is divided as far as pectoralis major which may be entered in a muscle splitting fashion, or divided with cautery. Occasionally it may be necessary to divide or resect part of a rib to obtain better access. This incision is used for:

- Upper hemisternotomy: aortic valve surgery, thymectomy.
- Lower hemisternotomy: mitral and tricuspid valve surgery, ASD.

Subxiphoid incision

This approach is described on 📖 p490, and involves a 4–6cm incision extending from the xiphoid process which is divided, caudally (Fig. 10.1c). The pericardium is entered above the diaphragm. This approach allows access to:

- The PDA and distal OM vessels if extended caudally.
- Pericardial window.
- Access in reoperative cases for epicardial ablation.

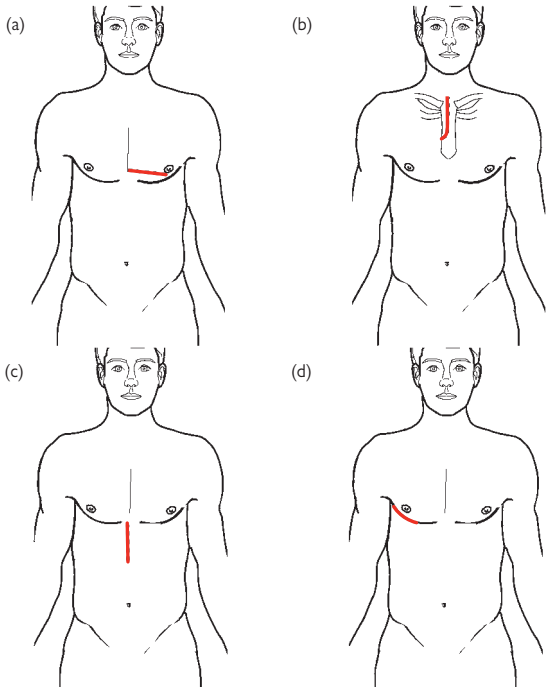


Fig. 10.1 (a) Left anterior minithoracotomy (MIDCAB), (b) hemisternotomy (AVR), (c) subxiphoid approach (pericardial window), (d) Right anterolateral minithoracotomy (mitral, tricuspid, ASD closure).

Thorascopic approaches

A modification of the left VATS approach described for thoracic procedures, requiring double lumen intubation and single lung ventilation (📖 pp686–687) can be used to harvest the LIMA and RIMA, and perform distal coronary anastomoses (usually directly via a MIDCAB incision). The main differences in set up are:

- External defibrillator paddles are placed.
- The radial arterial line should be sited on the right side.
- The patient is placed at a 30° angle (Fig. 10.3) with the hips horizontal and the left arm above the head.
- The patient is prepped so that the groin and the midline are accessible.
- The port sites are positioned as described on 📖 pp460–461.
- CO_2 is infused at 6–8mmHg by a pressure-regulated insufflator.
- A Harmonic scalpel is used by most surgeons to harvest the LIMA as there is less smoke, and local heat injury is less.

CPB through small incisions

Key facts

- Standard central cannulation can be modified for smaller incisions without majorly compromising safety or efficacy.
- Arterial and venous cannulation, clamping, venting, and delivery of cardioplegia can each be done via ***peripheral or central access***.
- The primary benefit is cosmetic: observational data shows mixed benefits in terms of pain control, recovery, blood loss, or extubation, and there is some evidence to suggest disadvantages include longer bypass times, lower mitral repair rates, and under-revascularization.
- Femoral arterial cannulation and retrograde flow is associated with an ***increased stroke risk*** in older patients and vasculopaths, who should always be identified by CT angiography if this strategy is considered.

Set-up for minithoracotomy mitral

Standard aorto-bicaval cannulation and cardioplegia delivery

In many patients a right 4th intercostal minithoracotomy (📖 p456) allows reasonable access to the aorta to cannulate: use the Fem-Flex Seldinger cannula as it is not possible to get your hand and a standard cannula in the incision. Better exposure of the aorta is obtained by carefully hitching up the pericardium and gentle caudal retraction on the aorta. The SVC and IVC are easily cannulated through this incision using standard single-stage cannulas, either directly or via the right atrial appendage. If vacuum assist is used, smaller cannulas will still provide excellent drainage. Retrograde cardioplegia catheters can be easily placed in the standard fashion. The antegrade cardioplegia cannula is most easily placed once on bypass. A Chitwood or flexible snake clamp is used to clamp the aorta. Lung isolation is needed for central cannulation: otherwise femoral cannulation allows institution of bypass and ventilation to be discontinued prior to thoracotomy.

Peripheral alternatives

Any of these can be used to improve access or reduce incision size:

- Femoral arterial cannulation (avoid in elderly or arteriopathic patients).
- Femoral venous cannulation (percutaneous or open) usually with vacuum assist to enhance venous drainage.
- Percutaneous internal jugular vein cannulation (anesthesiology).
- Percutaneous internal jugular retrograde cardioplegia cannulation.
- Fibrillatory rather than cardioplegic arrest (if no more than trace AI).

Port-access bypass

This involves fem-fem bypass, but also occludes the ascending aorta with a balloon (the ***endoclamp***) inserted via the femoral artery allowing delivery of antegrade root cardioplegia. The heart can be arrested on bypass, enabling mitral valve surgery, as well as CABG, to be carried out through smaller, usually thoracotomy or thorascopic (port), incisions. Aortic surgery is not possible because of difficulty maintaining the balloon occlusion distal to the aortotomy without occluding the head and neck vessels. Aortic dissection complicates around 1% of port access CPB.

Technique

As the surgeon is not able to visualize the cannulas or cardioplegia delivery system additional monitoring is mandatory. Radial arterial lines are sited bilaterally, and TEE routinely used so that migration of the arterial cannula or cardioplegia delivery system can be detected.

- Make a 2–3cm groin crease incision overlying the femoral vessels.
- Obtain proximal and distal control of the vessels with slings.
- After heparinization pass a 22F venous cannula into the RA via the femoral vein under TEE control, so the tip is in the SVC.
- Using vacuum assist enhances venous drainage.
- Cannulate the femoral artery with a dual port arterial return catheter and position the tip of the cannula in the aortic arch using TEE.
- Again under TEE pass the endoclamp down the second limb of the femoral cannula so that the tip sits just above the sinotubular junction.
- The endoclamp consists of a triple lumen catheter with an inflatable balloon at the tip: the balloon is inflated to occlude the aorta via the first lumen, the second allows aortic root pressure to be transduced and is used as a root vent, and the third delivers cardioplegia (Fig. 10.2).
- After institution of CPB inflate the endoclamp with 20–30mL of saline, aiming for a balloon pressure of 250–350mmHg. De-airing is critical.
- If there is any forward ejection of the heart the balloon will migrate distally, so it is important to ensure complete emptying of the heart.
- It is possible to puncture the balloon (e.g., while placing anterior mitral annuloplasty sutures): completing the procedure under fibrillatory arrest or replacing the balloon are both options.

Contraindications to port-access bypass

- Severe peripheral vascular disease.
- Intraluminal atherosclerosis of the aortic arch.
- Aortic valve or ascending aorta procedures.

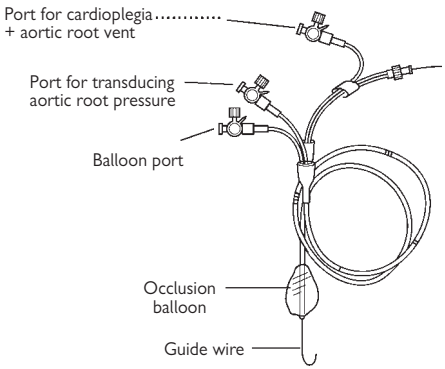


Fig. 10.2 Port-access endoclamp.

LIMA harvest via small incisions

Key facts

The LIMA can be harvested to a variable extent through several incisions. It is important to obtain a long length (preferably the entire length) of LIMA because:

- If the LIMA is tethered to the chest wall it will tend to kink.
- If the LIMA is short it will place the anastomosis under tension.
- A short length of LIMA will not reach the LAD in patients with distal disease or COPD.
- A long length facilitates sequential grafting.
- The risk of coronary steal from patent IMA branches is avoided.

Anterior, lateral, and mid-lateral MIDCAB incisions

There are a number of specialized retractors which enable the LIMA to be harvested from 1st to 5th rib space from the right of the patient in a manner similar to that used in a median sternotomy approach (Fig. 10.3).

Thoracopic IMA takedown

This is an alternative method of harvesting the IMA which has several advantages. Firstly, rib resection and retraction are avoided. Secondly, the entire length of the IMA can be mobilized, avoiding kinking, tension on the anastomosis, and IMA steal. Harvest takes longer, with a longer learning curve. Double lumen intubation and single lung ventilation is required, which not all patients tolerate well. Insufflation of CO_2 may result in hemodynamic compromise. Uncontrollable hemorrhage may mandate median sternotomy.

Patient selection

The following are relative contraindications to thoracopic LIMA harvest:

- Lung disease making single lung ventilation impossible.
- Previous thoracic surgery resulting in pleural adhesions.

Anesthesia and positioning the patient

This is described on 📖 p686. The patient's left armpit should be shaved. It may be helpful to mark the anterior and midaxillary lines as well as the 4th rib space and midline with indelible pen prior to prepping.

Positioning the ports

- Triangulate the port sites, which prevents the instruments from interfering with each other during use.
- Make three incisions of 1.5mm with a 15 blade.
- Use a mosquito or Dunhill to dissect down to the intercostal space.
- Ask the anesthetist to deflate the lung.
- Place the camera port at the 5th intercostal space along the anterior axillary line, which gives the best view of the length of the LIMA.
- The two instrument ports are placed using the camera internally to obtain optimal positioning, usually along the midaxillary line, one or two rib spaces either side of the camera port.
- Placing the instrument ports below and away from the LIMA gives better access for harvesting than placing them too close or directly opposite.

Harvesting the LIMA thoracoscopically

Filmy adhesions can usually be divided without wasting too much time, but dense adhesions make introduction of ports and instruments hazardous, and the thorascopic approach should be abandoned. If the LIMA is not obvious, it is easiest to identify it just distal to its origin from the subclavian artery where it is covered by a thin layer of parietal pleura. The middle 1/3 is frequently covered by fat, and the distal 1/3 by the innermost intercostals and pericardial fat. The camera can be held either by an assistant or by a voice-activated arm.

- Incise the parietal pleura lateral to the LIMA with cautery.
- Retract the pleura with forceps and use a combination of cautery and blunt dissection with the blade of the diathermy to separate the LIMA pedicle from the thoracic wall.
- LIMA branches are best divided by direct coagulation as clipping individual branches means frequent instrument changes, increasing operative time and the risk to adjacent structures.
- Some surgeons advocate skeletonizing the LIMA as a more precise dissection and longer length is allowed: this is at the risk of increased trauma to the LIMA.
- Insufflation of CO_2 to a pressure of 5–15mmHg, if tolerated by the patient, enlarges the operative field facilitating the distal dissection.
- Once an adequate length is secured heparin is given (150U/kg).
- The LIMA is ligated with two large clips and all but a few muscle strands divided, so that it is held in the anatomical orientation while the thoracotomy incision is made, through which it can be brought under direct vision.

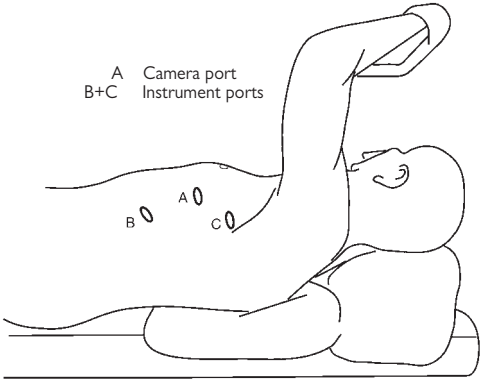


Fig. 10.3 Positioning the patient and port sites.

SVG harvest via small incisions

Key facts

- In addition to significant postoperative pain and decreased mobility, the standard method of harvesting the saphenous vein through a long, continuous incision results in delayed wound healing due to cellulitis, lymphoceles, edema, large skin flaps, fat necrosis, hematoma, sympathetic dystrophy.
- Much of this can be avoided by using either endoscopic or hybrid approaches which maintain skin bridges.
- The advantage of the standard open approach is that the 'no touch' technique preserves endothelial integrity: minimally invasive approaches result in much more direct handling of the vein and there is some observational data suggesting this may be associated with reduced long-term patency rates.

Methods of minimally invasive SVG harvest

There are several systems for minimal access harvest. In skilled hands all of them result in comparable quality conduit. Endoscopic techniques have a much longer learning curve: they are therefore better suited to surgical assistants on long-term contracts than surgeons on short training rotations. An alternative is using a stripper, which allows vein to be harvested through multiple small stab incisions without a camera.

Using a stripper (Fig. 10.4a)

A stripper is simply a 1cm metal ring, mounted at an angle on a long handle. This method is quicker than conventional harvest as there is no long incision to close, but real care should be taken to avoid avulsing small branches and damaging the endothelium with too much traction.

- Expose the saphenous vein at the groin and ligate it proximally.
- Place a 0/0 silk suture distally and cut it long.
- Pass the suture and the vein through the ring end of the stripper and pass the stripper down the vein until you meet resistance with skin tethering causes by a branch.
- Make a 1.5cm incision down onto the stripper where it has come up against a branch, and pass the stripper and vein up through the incision.
- Remove the vein from the stripper and the incision, and ligate and divide the branch is usually easily visible.
- Continue as far as necessary and test the vein in the usual manner.

Endoscopic vein harvest (Fig. 10.4b)

There are several systems available, and they offer different approaches to the various aspects of endoscopic vein harvest:

- The saphenous vein is identified just above the knee.
- The subcutaneous space is created by blunt dissection anterior to the vein either by a transparent blunt-tipped dissector under endoscopic vision, with an airtight space maintained by a balloon inflated at the incision and CO_2 insufflation: excellent vision is provided by the scope.

- Dissection is carried out with long-handled, blunt 'U' tipped instruments: side branches are clipped and divided with diathermy.
- Similar techniques have been used for radial artery harvest.
- Experienced surgical assistants take no longer to extract the length of vein in this way than by the open technique: the main issue is not one of time but one of quality, and the learning curve is around 200 cases.

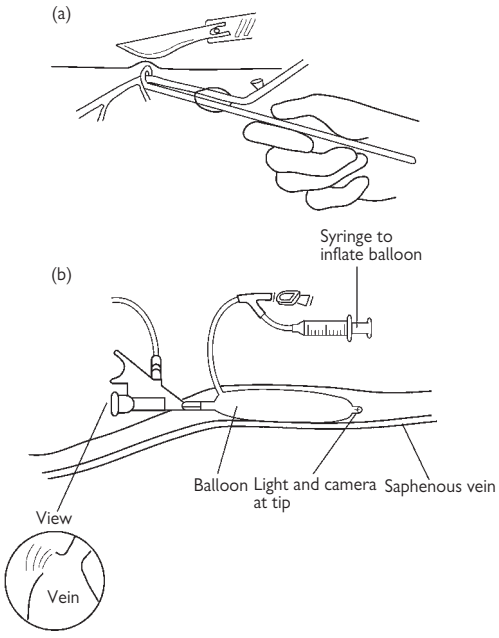


Fig. 10.4 (a) Using a vein stripper. (b) Endoscopic vein harvest.

CABG via small incisions

There are several methods of carrying out minimally invasive CABG. Four representative methods are described briefly in Box 10.1, and the techniques, their advantages and disadvantages listed in Table 10.1.

Box 10.1 Techniques of minimally invasive CABG

MIDCAB (minimally invasive direct coronary artery bypass grafting)

- LIMA harvesting and coronary artery anastomosis are performed, without bypass, through a small anterior thoracotomy incision.

EndoACAB (endoscopic atraumatic coronary artery bypass grafts)

- LIMA is harvested thorascopically, then coronary anastomosis is performed, off-pump, through a small anterior thoracotomy incision.

Port-access CABG (port-access coronary artery bypass grafting)

- LIMA harvesting, and coronary artery anastomosis on bypass, are performed through a small anterior thoracotomy, aortic occlusion and delivery of cardioplegia are endovascular.

TECAB (totally endoscopic coronary artery bypass grafting)

- LIMA harvesting and coronary artery anastomosis are performed, off bypass, thorascopically: the coronary artery anastomosis is usually performed with robotic assistance, and/or anastomotic aids (Fig. 10.5).

Table 10.1 Comparison of four methods of minimally invasive CABG

	MIDCAB	EndoACAB	Port-access	TECAB
Single lung ventilation	No	Yes	No	Yes
TEE/X-ray guidance	No	No	Yes	No
Arterial monitoring	R radial	R radial	R+ L radial	R radial
Femoral cannulation	No	No	Yes	No
Beating heart	Yes	Yes	No	Yes
Cardioplegia	No	No	Yes	No
Heparinization	Partial	Partial	Full	Partial
Endoscope required	No	Yes	No	Yes
Robot required	No	No	No	Yes
Thoractotomy incision	8–10cm	6–8cm	8–10cm	No
Ribs divided/excised	Yes	No	Yes	No
LIMA to LAD possible	Yes	Yes	Yes	Yes
Grafts to PDA, OM	No	Yes	No	Yes
Top-ends possible	No	Yes	No	Yes
Anastomotic devices	Not routinely used but helpful in TECAB (Fig. 10.5)			
Stenting some lesions	Inaccessible lesions stented in 'hybrid' procedures			

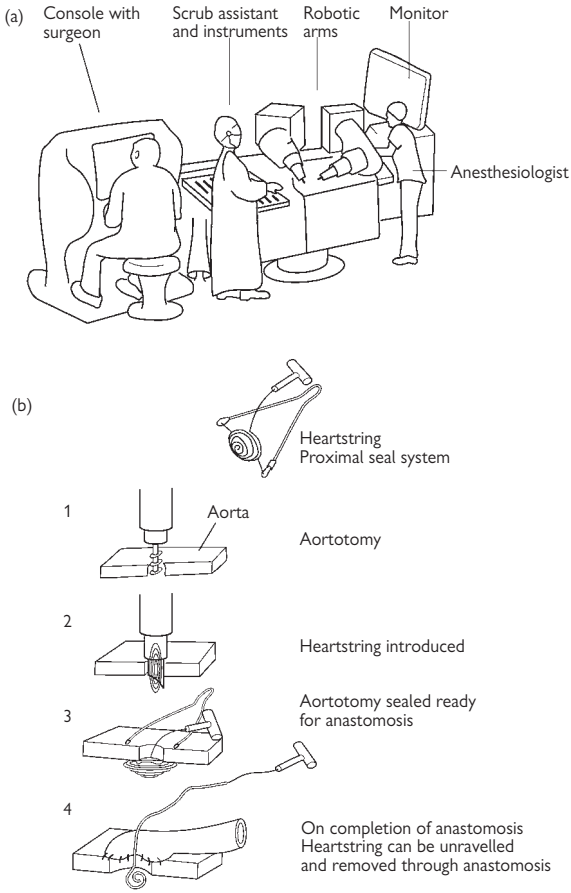


Fig. 10.5 (a) Set up for robotic surgery. (b) Device for performing proximal coronary anastomosis without using a clamp on the aorta.

Valve surgery via small incisions

Mitral surgery via right minithoracotomy

The main advantage of this approach is cosmetic. There is mixed observational data regarding possible benefits such as reduced transfusion requirement, earlier discharge, and quicker recovery. Potential disadvantages include a higher incidence of pulmonary complications, and stroke related to retrograde arterial perfusion in arteriopaths.

Preoperative selection and preparation

This approach requires either single lung ventilation or peripheral cannulation. Port-access bypass should be monitored via bilateral radial arterial lines. A TEE is mandatory to assess valve and ventricular function and de-airing. The patient is positioned with the right chest elevated by 30–45°, shoulders tilted back, and right arm either suspended above the head or well behind the posterior axillary line (Fig. 10.6). Eternal defibrillator pads should be placed. Expose both groins.

Technique of surgery

- Set-up for aorto-caval and fem-fem bypass via this approach is described on 📖 p458. Port-access cannulation is described on 📖 p458.
- The right 4th rib is exposed by a 6–8cm submammary skin incision, divided, and a mini-McCarthy/modified Cosgrove retractor inserted.
- The right lung is deflated, the pericardium incised 2cm anterior to the phrenic nerve under direct vision as far as the aortic reflection, and the posterior pericardial edge hitched firmly to the skin edge rotating the heart so that the left atrium is uppermost (Fig. 10.6), and the anterior pericardium hitched to the chest wall to expose the aorta.
- A port and 0° camera can be placed through the 3rd rib space.
- If port-access bypass is not being used, an aortic root vent and cardioplegia catheter can be placed directly (easiest in the decompressed heart on bypass), and the aorta cross-clamped by inserting the cross-clamp through a 5mm incision in the 3rd intercostal space (Fig. 10.6m) or directly with a flexible clamp.
- Fibrillation is an alternative to cardioplegia (see 📖 p121 for principles).
- Sondergaard's groove is dissected for 1–2cm, a small atriotomy is made medial to the right superior pulmonary vein, and a purpose designed retractor inserted to expose the mitral valve (Fig. 10.6).
- Endoscopic instruments are used. Knot-tiers may not be necessary in slim patients. Tricuspid and maze procedures can be performed.
- The left atriotomy is closed under direct vision in the standard fashion.

Aortic valve surgery

The aortic valve can be accessed via an upper sternotomy incision (📖 p456) or a small right anterior thoracotomy. The technique is a modified version of the standard open technique, although port-access bypass can be used instead of aorto-caval bypass. Retrograde cardioplegia is used if desired, and the heart is vented via the right superior pulmonary vein, or LV via the dome of the LA. De-airing is carried out under TEE guidance in the standard fashion (📖 p114), but lifting the heart is done using internal defibrillator paddles.

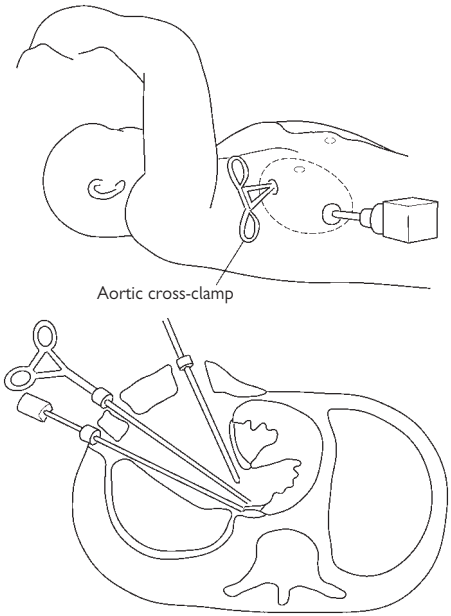


Fig. 10.6 Minimal access mitral valve surgery.

Percutaneous valve options

Key facts

- Pericardial valves (CoreValve™ and Sapien™) mounted on expandable stents are used to perform percutaneous aortic valve replacement, currently in patients with calcific aortic stenosis considered high- or very high-risk surgical candidates.
- Access is via the femoral artery, although transapical (left anterior thoracotomy) and axillary artery via cut-down are also used.
- The MitraClip is the most widely used option for mitral repair: results are less comparable to surgery, and indications more limited.

Corevalve™

Device

Porcine pericardium mounted on a self-expanding, large waisted nitinol stent in 22mm and 26mm sizes, now delivered via 18F sheaths (down from earlier 25F and 21F), designed to sit across the LVOT and sinuses (Fig. 10.7a).

Indications

- High-risk surgical candidate (predicted operative mortality ≥15%).
- Symptomatic, severe, degenerative AS.

Contraindications

- Native aortic annulus <20mm or >27mm, mixed aortic valve disease (AI >3–4+), severe septal hypertrophy with LVOT gradient, bicuspid aortic valve, ascending aorta >43mm.
- Femoral arteries unable to accommodate 18F sheath.
- Aortic root angulation (angle between plane of aortic valve annulus and horizontal plane/vertebrae) >70° (femoral access) or >30° (axillary access).

Results

- Procedural success rate close to >90% in most recent series.
- 30-day mortality 0–25%, stroke (0–10%), conversion to surgery (0–8%).
- Need for PPM insertion (0–36%).
- Complications include paravalvular leak >2+ (4–35%), tamponade 2–10%), malpositioning, aortic dissection (0–4%), blood transfusion 2U (3–24%), vascular complications (10%), 30-day MACCE 3–30%.
- Procedure duration 2–3h, length of stay 7–17 days.

Sapien valve

Device

Bovine pericardium mounted in a balloon expandable 16mm long tubular stent in 26mm sizes, now delivered via 22F sheaths designed to sit across the annulus (Fig. 10.7b).

Indications

- High-risk surgical candidate (predicted operative mortality ≥15%).
- Symptomatic, severe, degenerative AS.

Contraindications

- Native aortic annulus <18mm or >25mm on echo.
- Mixed aortic valve disease (AI >3+) or severe MR.

- Severe septal hypertrophy with LVOT gradient.
- Bicuspid or non-calcified aortic valve, active endocarditis, bulky calcified aortic valve leaflets in close proximity to coronary ostia.
- Significant abdominal or thoracic aortic disease, e.g., aneurysm (>5cm), marked tortuosity, aortic arch atheroma, narrowing of abdominal aorta with calcification or irregularities, ileofemoral calcification, or vessels <7mm for 22F catheter.
- Femoral arteries unable to accommodate 18F sheath.
- Aortic root angulation (angle between plane of aortic valve annulus and horizontal plane/vertebrae) >70° (femoral access) or >30° (axillary access).

Results

- PARTNERS trial[1] randomized to transcatheter (n=348) or surgical replacement (n=351).
- Procedural success rate 95%.
- 30-day mortality 3.4% Sapien vs. 6.5% surgery (p=0.07), 1-year mortality 24 vs. 27% (p=0.44).
- Strokes: Sapien 5.5% vs. surgery 2.4% (p=0.02) and at 1 year 8.4 vs. 4.3%.
- 12% moderate AI in Sapien (vs. 0.9% after surgery).
- Major vascular complications 11% in Sapien vs. 3% in surgical group (p < 0.001), major bleeding 9 vs. 19.5% (p=0.006).
- Shorter hospitalization with Sapien (6 vs. 12 days) no functional difference at 1 year.

MitraClip

Device

4mm wide, cobalt-chromium implant with two arms designed to clip the anterior and posterior mitral leaflet together (edge-to-edge repair, 📖 p410) delivered by atrial transseptal puncture (Fig. 10.7c).

Indications

- MR 3–4+, primary jet originating from malcoaptation of P2 and A2 (📖 p408).

Contraindications

- Mitral valve orifice <4.0cm^2, flail segment ≥15mm wide, flail gap ≥10mm, or coaptation depth >11mm/vertical coaptation depth <2mm if leaflet tethering, significant adjacent cleft, bileaflet flail.
- Severe mitral annular calcification, lack of chordal support, prior mitral surgery, endocarditis, rheumatic heart disease, ASD, or PFO.

Results

- At 1 year freedom from death, surgery, and grade 3 or 4+ MR was 55% in the MitraClip goup and 73% in the surgery group (p <0.001).
- Major adverse events at 30 days were the same for death (1%), stroke (2%), emergency cardiac surgery 2–4%, and the difference in adverse events reported was due to increased blood transfusion in the surgery arm.

Reference

1. Smith CR *et al.* Transcatheter vethus surgical aortic-valve replacement in high-risk patients. *New Engl J Med* 2011; **364**:2187–98.

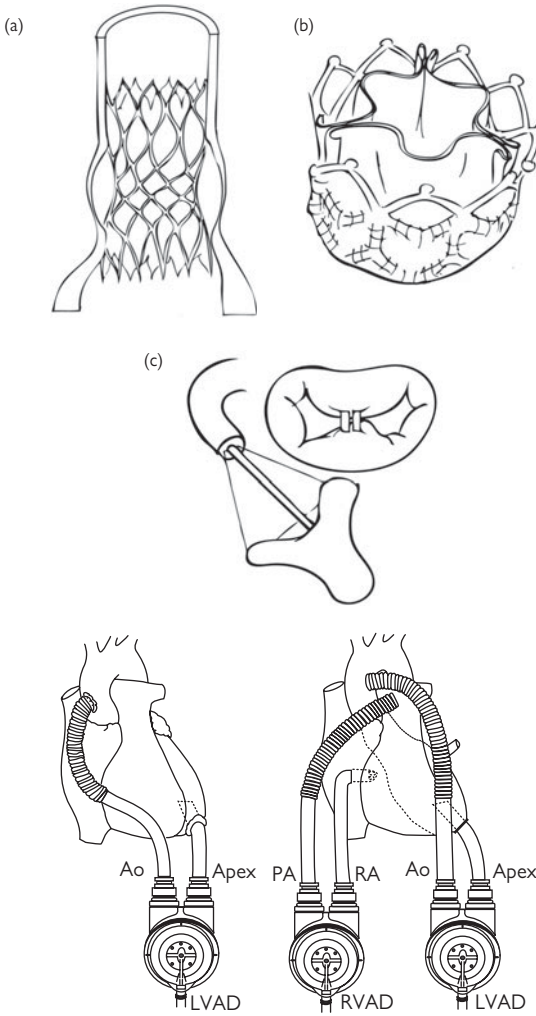


Fig. 10.7 (a) Corevalve aortic prosthesis. (b) Sapiens aortic prosthesis. (c) Mitraclip. Various assist devices: (Top right) Thoratec Heartmate VE. (Top left) Jarvik 2000 implanted with outflow in descending aorta. (Middle) Tandem and Impella devices. (Bottom left) schematic of implantation of Thoratec LVAD. (Bottom right) schematic of Thoratec biventricular device.